

CSE 5830 Homework 1

9/9/2026

Due: 11:59PM, Wednesday 9/16/2026

Show all your work to get a full score.

Problem 1-Prove (10 points)

$$p(x, y|z) = p(x|z)p(y|x, z)$$

and also

$$p(x|y, z) = \frac{p(y|x, z)p(x|z)}{p(y|z)}$$

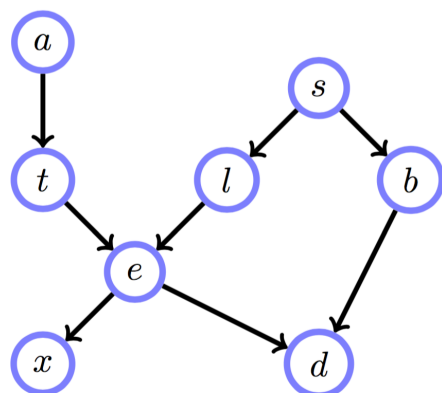
Problem 2- A college has three departments, A, B, and C. The percentage of female students in departments A, B, and C are 35%, 20% and 55%, respectively. A department is chosen at random, and then a student is selected randomly from that department. (15 points)

- a) What is the probability that the selected student is female?
- b) Given that a female student is selected, what is the probability that she is from Department B?
- c) If a male student is selected, what is the probability that he is from department A?

Problem 3-Consider three variable distributions, $P(A, B, C, D)$ where random variables A and B can have two values (binary) and random variables C and D can have 3 values. (20 points)

- a) How many parameters are needed to specify this distribution?
- b) Assuming the distribution can be factorized as $P(A, B, C, D) = P(A)P(B|A)P(C|B)P(D|B)$. How many parameters are needed to specify distributions of this form?
- c) Draw a network that represent this distribution. Is $A \perp B$? is $A \perp B|C$?

Problem 4- Consider the Chest Clinic belief network in the following figure. (30 points)



x = Positive X-ray
 d = Dyspnea (Shortness of breath)
 e = Either Tuberculosis or Lung Cancer
 t = Tuberculosis
 l = Lung Cancer
 b = Bronchitis
 a = Visited Asia
 s = Smoker

The Chest Clinic network concerns the diagnosis of lung disease (tuberculosis, lung cancer, or both, or neither). In this model a visit to Asia is assumed to increase the probability of tuberculosis.

a) State if the following conditional independence relationships are true or false (4 points)

1. tuberculosis $\perp$ smoking | shortness of breath
2. lung cancer $\perp$ bronchitis | smoking
3. visit to Asia $\perp$ smoking | lung cancer
4. visit to Asia $\perp$ smoking | lung cancer, shortness of breath

b) Calculate the values for $p(d)$, $p(d|s = tr)$, $p(d|s = fa)$.

The table values are:

$p(a = tr)$	$= 0.01$	$p(s = tr)$	$= 0.5$
$p(t = tr a = tr)$	$= 0.05$	$p(t = tr a = fa)$	$= 0.01$
$p(l = tr s = tr)$	$= 0.1$	$p(l = tr s = fa)$	$= 0.01$
$p(b = tr s = tr)$	$= 0.6$	$p(b = tr s = fa)$	$= 0.3$
$p(x = tr e = tr)$	$= 0.98$	$p(x = tr e = fa)$	$= 0.05$
$p(d = tr e = tr, b = tr)$	$= 0.9$	$p(d = tr e = tr, b = fa)$	$= 0.7$
$p(d = tr e = fa, b = tr)$	$= 0.8$	$p(d = tr e = fa, b = fa)$	$= 0.1$

$p(e = tr|t, l) = 0$ only if both t and l are **fa**, 1 otherwise.

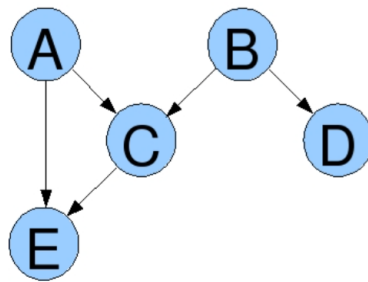
You can write a code for this problem. If you do that submit the code and the code output. I recommend to send .ipynb file and do not clear the output.

Problem 5- Consider the distribution: (15 points)

$$p(x, y, w, z) = p(z|w)p(w|x, y)p(x)p(y)$$

- Draw a network that represent this distribution. (3 points)
- Write $p(x|z)$ using a formula involving (all or some of) $p(z|w)$, $p(w|x, y)$, $p(x)$, $p(y)$. (4 points)
- Write $p(y|z)$ using a formula involving (all or some of) $p(z|w)$, $p(w|x, y)$, $p(x)$, $p(y)$. (4 points)
- Using the above results, derive an explicit condition for $x \perp y | z$ and explain if this is satisfied for this distribution. (4 points)

Problem 6- Consider the following graph. (15 points)



- Which pairs of variables are independent in the graphical model above, given that none of them have been observed? (4 points)
- Assume that the value of E is known. (E is observed. A, B, C, and D are not observed.) Which pairs of variables (not including E) are independent in the same graphical model, given E? (3 points)
- What is the decomposition of the joint distribution $P(A, B, C, E)$? (4 points)
- How many independent parameters are required to uniquely define the **CPD of E** (the conditional probability distribution associated with the variable E) if A, B, and D are binary, and C and E have three values each? (4 points)